



Addis Ababa University
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SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY!



ORAL CANCER

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Outline

- Introduction
- Epidemiology
- Risk factors
- Oral potentially malignant disorders
- Sub sites of the oral cavity
- Diagnosis and work up

Introduction

- The functionality of the oral cavity is unmatched in the human body
 - Speech, mastication,
 - taste, swallowing
 - respiration, and facial expressions
- Cancers of the oral cavity can greatly diminish these basic functions .
- In some cases, eliminate all together, which has devastating effects on quality of life.

Epidemiology

- Approximately 85 to 95% of all oral cancer is squamous cell carcinoma (SCC).
- Other minority of oral cavity ca represents ~ **10 %**
 - Sarcomas
 - melanomas
 - salivary gland tumors
 - lymphomas & metastatic diseases from other sites
- Oral cancer is the 11th most common malignancy in the world.
- Gender ?
- Projections ?

Cont.

- Approximately 50% of patients diagnosed with oral cancer will ultimately die of their disease.
- The American cancer society recommends a cancer related check up, including examination cancers of oral cavity , every 3 years for asymptomatic men and women aged 20 to 39 years and yearly for men and women aged 40 years and older.

Squamous cell carcinoma (SCC)

Risk Factors for SCC of the Oral Cavity:

- Tobacco-benzopyrenes, benzanthraces, and epoxides
- Alcohol
- Viral etiology (HPV).
- Gene alteration
- Poor oral hygiene/non healing wound
- Immunosuppression
 - Hiv- kaposi sarcoma
 - Immunosuppressive drugs
 - Transplant—250x

Precancerous lesion and conditions

Tobacco

- The risk of oral cancer associated with tobacco use is noted to be 2 to 12 times higher than in the nonsmoking population.
- Up to 95% of individuals with oral cancer will have a tobacco use history.
- The specific causative agents in tobacco include **benzopyrenes, benzantracenes, & epoxides.**
 - Cigarettes and cigars
 - Smokeless tobacco(dip/snuff/snus)
 - Water pipe hook/sheesha/shisha
 - Areca nut/betel quid
 - Electronic cigarettes (e-cigarettes)

Alcohol

- Alcohol has long been considered a **co-risk factor** in the development of OCSCC
- **Synergistically** act with tobacco by solubilizing its toxins and concentrating them in the **dependent portions** of the OC: the floor of mouth and lateral borders of the tongue in particular.
- Mechanism ?
- 1.3 fold higher for those who drinks
- 2.5 fold higher for those who drink heavily

Oncogenic viruses

- ❖ The HPV subtypes 16 and 18, similar to those causing cervical cancer, have been implicated.
- The presence of HPV in the oral cavity was associated with a 3.7 times greater chance of cancer development than in the non infected individual.

Genetic factors

- Gene alteration is responsible for carcinogenesis, but rather a host of altered genes contribute.
- The first step in carcinogenesis is the initiation phase
- ✓ **Genetic defect** is created by *environmental agents* such as tobacco/EtOH/viruses
- ✓ **Dysfunctional cancer regulatory genes** - including proto-oncogenes such as growth factors (FGF, PDGF, TGF α and β , NF κ B)
- ✓ **Inherited defects** - tumor suppressor genes p53 and retinoblastoma (RB) may also contain inherited defects which initiate carcinogenesis

Other risk factors?

- Discussion

Oral Potentially Malignant Disease

- Oral potentially malignant disorders (OPMDs) are conditions that **precede the onset** of cancers of the oral cavity .
- PRECANCEROUS LESIONS
- CONDITIONS
- ORAL LICHEN PLANUS
- ORAL LICHENOID REACTIONS
- ORAL LUPUS ERYTHEMATOSUS
- HEREDITARY CONDITIONS

Precancerous lesion

- **Precancerous lesion** is morphologically altered tissue in which the development of malignancy is more likely than with normal mucosa.
- Broadly classified as leukoplakia and erythroplakia.

Leukoplakia

- Leukoplakia is a white patch or plaque that cannot be characterized clinically or ascribed to any other pathologic disease.
- It cannot be scraped or rubbed off
- Generally asymptomatic
- Clinically appears as a white or off-white lesion that may be flat, slightly elevated, irregular, or smooth.

Cont.

- Has a strong male predilection.
- Diagnosed after the 4th decade of life
- The tongue, buccal mucosa, lower buccal vestibule and floor of mouth account for most oral cavity leukoplakia.
- Malignant transformation rate ~ 17 %

leukoplakia

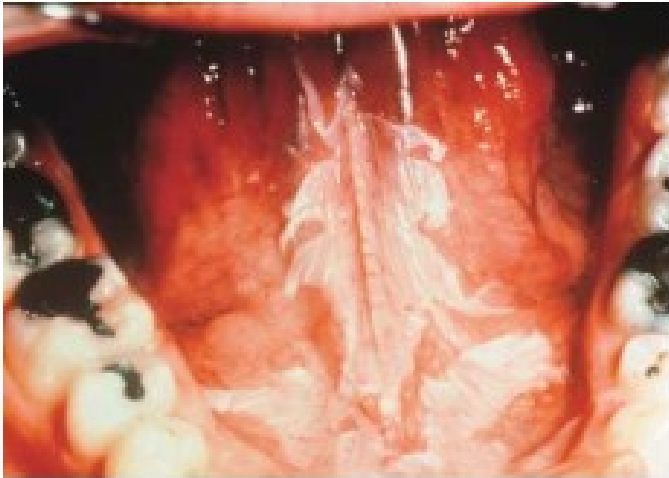


FIGURE 32-1 *Typical appearance of floor-of-mouth leukoplakia.*



Erythroplakia

is a red patch that cannot be scraped off or characterized clinically or ascribed to any other pathologic disease.

- The most common sites of occurrence are the palate, floor of the mouth and retromolar trigone.
- Lesions appear as bright red, are frequently “velvety” in appearance, and have a sharply demarcated border.



Erythroplakia

- Almost all true erythroplakia demonstrates dysplasia, carcinoma in situ, or invasive carcinoma.
- Mgt?
 - 9% - mild-to-moderate dysplasia
 - 40% - were carcinoma in situ or severe dysplasia
 - 51% - were invasive SCC

Mixed Lesions

- **Erythroleukoplakia** - are leukoplakias with a red component, which may indicate colonization of *Candida species*
- Erythroleukoplakia has a high incidence of associated dysplasia, carcinoma in situ, and frank carcinoma.
 - Malignant transformation rate as high as 23%



Precancerous condition

- **Precancerous condition** is a condition or generalized disease that does not necessarily alter the appearance of the mucosa but may be associated with a greater risk for the development of cancer.”
- Oral submucous fibrosis (OSF) is a precancerous condition most frequently associated with the habit of chewing betel quids.
- The condition is characterized by a mucosal rigidity that leads to trismus, odynophagia with spicy foods, and difficulty with speech and swallowing.



Clinical Correlation

Based on Site

- The boundaries of the oral cavity extend from the vermiliocutaneous junction of the lips to the junction of the hard and soft palate posterior-superiorly and to the line created by the circumvallate papilla posterior-inferiorly.
- Posterior-laterally the boundaries are represented by the anterior faucial pillars.
- The American Joint Committee on Cancer (AJCC) has divided the oral cavity into seven distinct anatomic locations from which primary lesions may develop.

Mucosal Lip

- The lip begins at the junction of the vermilion border with the skin and includes only the vermilion surface or that portion of the lip that comes into contact with the opposing lip.
- Mucosal lip cancer is seen almost exclusively in older white men as a result of chronic sun exposure.
- Nodal metastasis in lip cancer is infrequent.
- Metastasis from the lower lip is to the submental, submandibular, and perifacial nodes (level I more commonly than level II).
- Preauricular, periparotid, and submandibular nodes drain cancers of the upper lip and commissure (level II more commonly than level I).
- Bilateral neck metastasis may develop if the lower lip lesion is near or has crossed the midline; however, the upper lip rarely exhibits crossover between right- and left-side lymphatics.



FIGURE 32-7 *Neglected carcinoma of the lower lip.*

Buccal Mucosa

- ✓ Buccal mucosa includes all the lining of the inner surface of the cheeks and lips from the line of contact of the opposing lips (mucovermilion junction) to the line of attachment of mucosa to the alveolar ridge (upper and lower) and pterygo-mandibular raphe .
- ✓ First-echelon lymphatic drainage from the buccal mucosa is level I followed by level II.
- Carcinoma of the buccal mucosa represents 2 to 10% of all SCC of the oral cavity but In Central and Southeast Asia represents more than 40% of all oral cavity SCC.
- Cervical metastases are observed in 10 to 27% of presenting patients.

Buccal mucosa Ca



FIGURE 32-8 *Squamous cell carcinoma of the left buccal mucosa.*

Alveolar Ridge

- ✓ The alveolar ridge mucosa may be divided into lower (mandibular) and upper (maxillary) components.
- Alveolar ridge or gingival carcinoma represents 2 to 18% of oral cancers
- Occurs predominantly on the mandibular alveolus (64 to 76%).
- At diagnosis, approximately one-third of these tumors exhibit some bony involvement.
- ✓ Nodal drainage is principally to levels I and II for both the maxillary and mandibular lesions

Alveolar ridge or gingival ca



Retromolar Gingiva (Retromolar Trigone)

- The retromolar gingiva is a triangular region of attached mucosa overlying the ascending ramus of the mandible from the level of the posterior surface of the last molar tooth superiorly to the tuberosity of the maxilla.
- Primary symptomatic complaints with these tumors are Sore throat, otalgia, and trismus.
- ✓ Lymphatic drainage from this area is predominantly to the submandibular nodes (level IB) and the upper jugulodigastric nodes (level II).
- ✓ Lesions of this region tend to be more aggressive in nature with regard to developing cervical metastasis.



Floor of the Mouth

- The floor of the mouth is a semilunar space over the mylohyoid and hyoglossus muscles, extending from the inner surface of the lower alveolar ridge to the under surface of the tongue.
- Its posterior boundary is the base of the anterior faucial pillar of the tonsil.
- Studies have shown that nearly one-half of all patients presenting with a floor-of-mouth carcinoma will have metastatic disease at presentation.

Two distinct lymphatic drainage :

- ✓ The **superficial system** drains bilaterally into the submandibular nodes.
- ✓ The **deep system** drains into the ipsilateral submandibular , upper and middle jugulo digastric nodes (levels I, II, and III).

FOM



Hard palate

- The hard palate is between the upper alveolar ridge and the mucous membrane covering the palatine process of the maxillary bones.
- There is a paucity of lymphatics to the hard palate.
- Approximately 10 to 25% of individuals present with evidence of metastasis, generally to levels I and II.
- Hard palate lesions may also metastasize to retropharyngeal nodes or nodes that are not palpable on a clinical examination.
- Non healing ulcers and poor fitting dentures are common complaints.

Hard Palate Ca



Oral Tongue

- The anterior two-thirds of the tongue is the freely mobile portion that extends anteriorly from the line of circumvallate papillae to the undersurface of the tongue at the junction of the FOM.
- It has four areas: the tip, the lateral borders, the dorsum, and ventral surface.
- Lymphatic drainage of the oral tongue is principally to level II, followed by levels III and I.
- Tongue carcinoma represents 22 to 49% of all oral cancer Dx.
- Carcinoma of the lateral border generally metastasizes ipsilaterally, but SCC of the tip or body of the tongue may exhibit bilateral metastases.
- Approximately 40% of patients have evidence of clinical node metastasis at the time of diagnosis.

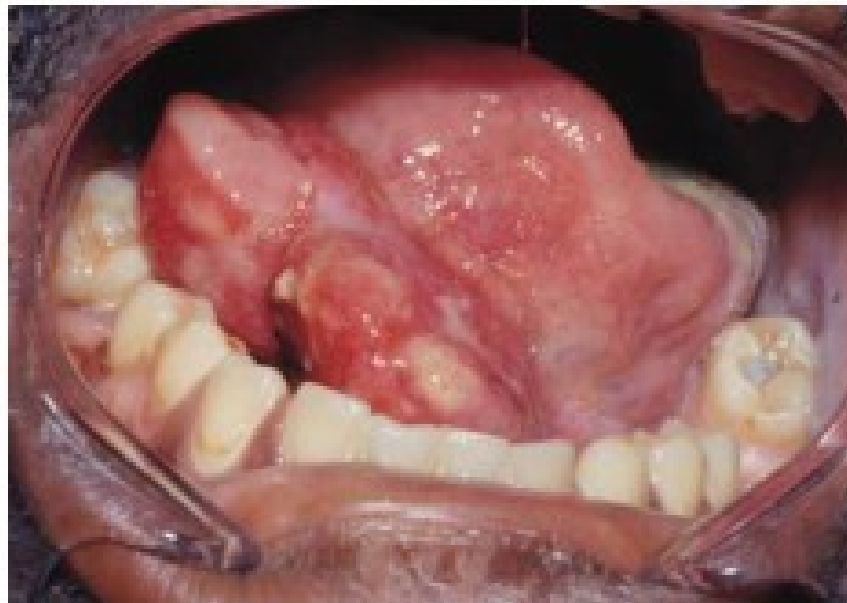


FIGURE 32-13 *Carcinoma proliferating from ventral tongue to encompass full thickness of the tongue.*

Assessment of Primary Lesion:

Complaints:

- non-healing sore in the mouth,
- loosening of teeth,
- ill-fitting dental prosthesis,
- trismus,
- otalgia, or
- weight loss.

Examination of the oral cavity should include:

- **use of a dental mirror** for indirect evaluation of the nasopharynx and hypopharynx.
- **Bimanual palpation** to assess any involvement of structures.
- **Anterior and lateral traction on the tongue** with cotton gauze.

The **growth patterns** of a tumor are **endophytic, exophytic, or ulcerated**.

Assessment of Regional Metastasis:

- Digital palpation of all levels of the neck bilaterally.
- Observation of the neck.
 - Palpate the neck standing behind the patient.
- Lymph node chains should be evaluated for the presence of palpable masses, their size, surgical neck level, and whether the mass is fixed or moveable.
- Other important palpable structures include the parotid gland, the thyroid gland, and the postauricular, occipital, and supraclavicular lymph node chains.
- The parotid gland should be evaluated for the presence of any palpable disease or masses and the thyroid gland for any nodule, masses, or thyromegaly.
- The trachea should be inspected for any deviation or fixation.

Computed Tomography:

- ✓ A lymph node is considered abnormal when it is greater than 1.5 cm in the jugulodigastric region or greater than 1 cm in other regions of the neck , has a round or sphere-like shape ,and specifically when intranodal necrosis is present, independent of size and shape.

Magnetic Resonance Imaging:

- ✓ When using MRI for evaluating the neck the same criteria concerning nodal size, shape, and central necrosis should be applied as when evaluating with CT.

Ultrasound:

- ✓ Criteria for the evaluation of potentially malignant cervical nodes with sonography also involve the assessment of nodal size, shape, and presence of central necrosis.
- ✓ Metastatic nodes are frequently hypoechogenic . In the presence of extracapsular spread, loss of border definition is observed. Normal lymph nodes are frequently difficult to detect.

Assessment of Distant Metastasis:

- ✓ Distant metastasis from the oral cavity most frequently involves the lung, followed by liver and bone.
- ✓ Therefore, routine posterior-anterior and lateral chest radiographs and the evaluation of liver function tests (LFTs) are considered the minimum metastatic work-up for head and neck cancer patients.

Prognosis

- ✓ A thorough clinical examination is the first line of defense in the detection of oral cancer. Prognosis is directly dependent on the tumor stage at diagnosis.

Biopsy:

- Incisional
- Excisional (for small lesions),
- Brush cytology is a safe, minimally invasive technique for use in the screening of clinically suspicious lesions.

Management

- Surgery
 - ❖ For most stage I and stage II cancers of the oral cavity, surgical resection with Frozen Section Analysis of the margins is advocated by most clinicians.
 - ❖ Surgical excision of the lesion plus neck dissection
- Radiotherapy
 - ❖ Two or more lymph nodes containing metastatic disease in a neck dissection (many clinicians contend that one positive node is an indication)
 - ❖ Extracapsular extension (ECS) of cancer beyond the confines of a node
 - ❖ Poor histologic factors: extensive perineural or perivascular invasion, positive (close) soft tissue margins
 - ❖ Large (T3 or T4) primary cancers
- Chemotherapy
 - ❖ Currently the role for chemotherapy in oral cavity cancer is limited to use in unresectable disease in which it is combined with radiation treatment, metastatic disease, or recurrence.

staging

- The TNM system devised by the 8th edition of AJCC is designed to stratify cancer patients into different stages based on the characteristics of the primary tumor (T) and depth, regional lymph node metastasis (N), and distant metastasis (M).
- It is an attempt to help guide treatment and estimate patients' 5-year survivability.
- *T refers to the primary* lesion and is graded on greatest dimension and presence of adjacent tissue infiltration (Table 32-1).

staging

- *N* refers to regional lymph node involvement and is graded on the presence of nodes, greatest dimension, and side of involvement in relation to the primary tumor (Table 32-2).
- *M* grades distant metastasis and is based simply on its presence (M1) or absence (M0).
- The AJCC staging system is designed for clinical use; however, the patient may be restaged based on final pathology after resection and designated with a *p* prefix (pTNM) or at autopsy with an *a* (*a*TNM).
- If synchronous tumors are found at presentation, the higher stage tumor should be used for stage designation, and an *m* suffix may be used to denote the multiple primary tumors (T_mNM).

Table 32-1 Primary Tumors (T)

<i>Tumor</i>	<i>Description</i>
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ
T1	Tumor 2 cm or less in greatest dimension
T2	Tumor more than 2 cm but not more than 4 cm in greatest dimension
T3	Tumor more than 4 cm in greatest dimension
T4a*	Tumor invades adjacent structures (eg, through cortical bone, into deep [extrinsic] muscle of the tongue, maxillary sinus, skin of face) (resectable)
T4b	Tumor invades masticator space, pterygoid plates, or skull base or encases internal carotid artery (unresectable)
*Superficial erosion alone of bone or tooth socket by an alveolar primary is not sufficient to classify a tumor as T4. Adapted from Greene FL et al. ^{35,36}	

Table 32-2 Regional Lymph Nodes (N)

<i>Node</i>	<i>Description</i>
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in a single ipsilateral lymph node, 3 cm or less in greatest dimension
N2	Metastasis in a single ipsilateral lymph node, more than 3 cm but not more than 6 cm in greatest dimension; or in multiple ipsilateral lymph nodes, none more than 6 cm in greatest dimension; or in bilateral or contralateral lymph nodes, none more than 6 cm in greatest dimension
N2a	Metastasis in a single ipsilateral lymph node more than 3 cm but not more than 6 cm in greatest dimension
N2b	Metastasis in multiple ipsilateral lymph nodes, none more than 6 cm in greatest dimension
N2c	Metastasis in bilateral or contralateral lymph nodes, none more than 6 cm in greatest dimension
N3	Metastasis in a lymph node more than 6 cm in greatest dimension
Adapted from Greene FL et al. ^{35,36}	

The neck is divided into six “surgical levels based on anatomic structures:

Level I

- Includes the submental and submandibular nodal groups.

Level II

- Includes upper jugular lymph nodes surrounding the internal jugular vein and adjacent spinal accessory nerve.

Level III

- Includes middle jugular lymph nodes surrounding the internal jugular vein.

Level IV

- Includes the lower jugular lymph nodes surrounding the internal jugular vein.

Level V

- Includes all the nodes in the posterior triangle, the spinal accessory and transverse cervical nodes, and all of the upper, middle, and lower jugular lymph nodes on the posterior aspect of the SCM.

Level VI

- Includes the pretracheal, paratracheal, and prelaryngeal or so-called Delphian lymph nodes.

levels

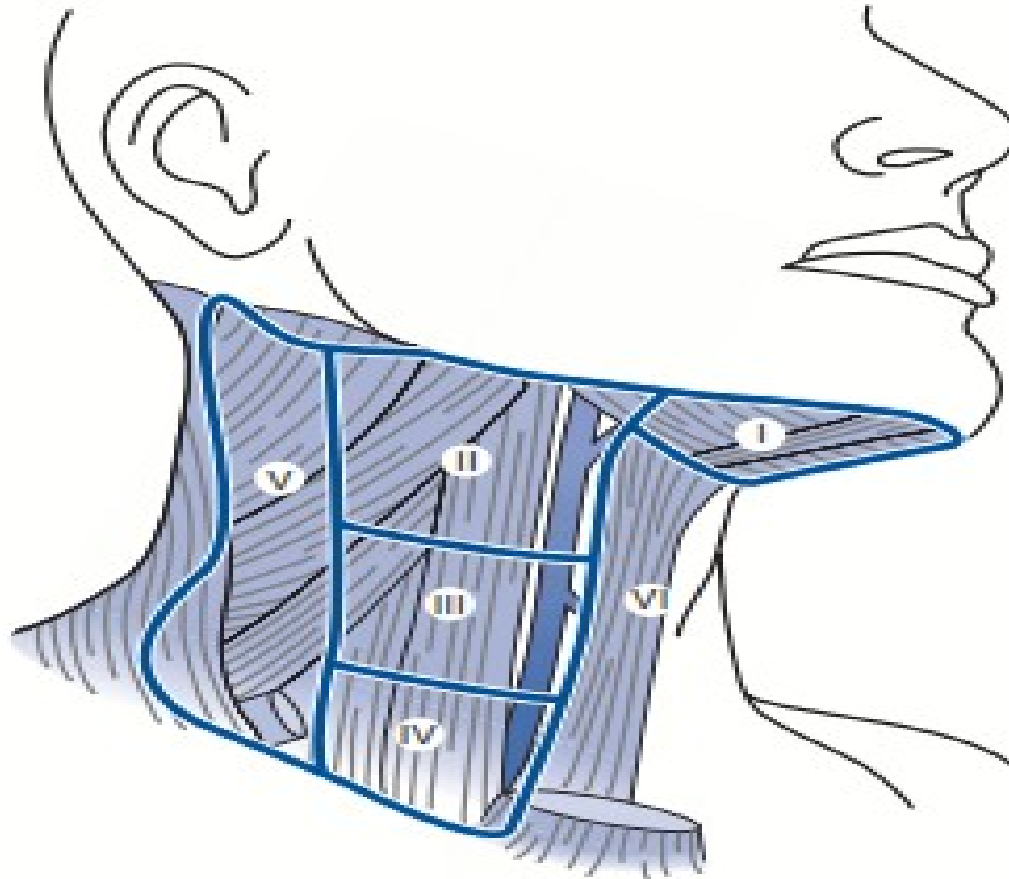


FIGURE 32-6 *Lymph node levels of the neck. Levels I to VI are subdivided and described in text.*

Table 32-3 Stage Grouping			
Stage	Characteristics		
Stage 0	Tis	N0	M0
Stage I	T1	N0	M0
Stage II	T2	N0	M0
Stage III	T3	N0	M0
	T1	N1	M0
	T2	N1	M0
	T3	N1	M0
Stage IVA	T4a	N0	M0
	T4a	N1	M0
	T1	N2	M0
	T2	N2	M0
	T3	N2	M0
	T4a	N2	M0
Stage IVB	Any T	N3	M0
	T4b	Any N	M0
Stage IVC	Any T	Any N	M1
Adapted from Greene FL et al. ^{35,36}			

Reading Assignment

- Read the 8th AJCC OCSCC staging
- Read about synchronous and metachronous tumor
- Read about the surgical/anatomic landmarks of the six cervical lymph node levels

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